

Unit 8

Eigenvalues and Eigenvector Bases

1. Eigenspaces

We first work with real vectors and real linear maps. In Section 4, we also calculate nonreal eigenvalues and eigenvectors of real matrices. Let $T : V \rightarrow V$ be linear. An *eigenvector* is a nonzero vector v for which $Tv = \lambda v$; the scalar λ is its *eigenvalue*. It is unique for that vector: if $\lambda v = \mu v$ and $v \neq 0$, then $\lambda = \mu$. The zero vector is not called an eigenvector. The *eigenspace*

$$E_\lambda(T) = \ker(T - \lambda I) = \{v : Tv = \lambda v\}$$

does include zero. Linearity gives $T(av + bw) = \lambda(av + bw)$ for $v, w \in E_\lambda$; therefore E_λ is a subspace, and every nonzero element of it is an eigenvector. A linear combination of eigenvectors may be zero.

Example 8.1. For $D = \text{diag}(2, 2, -1)$,

$$E_2(D) = \{(x, y, 0)^\top : x, y \in \mathbb{R}\}, \quad E_{-1}(D) = \{(0, 0, z)^\top : z \in \mathbb{R}\}.$$

There are infinitely many eigenvectors, two eigenvalues, and a basis of three eigenvectors. These are three different counts.

On the space of infinitely differentiable functions, differentiation satisfies $\frac{d}{dt}e^{\lambda t} = \lambda e^{\lambda t}$. Thus $e^{\lambda t}$ is an eigenvector of a linear map whose vectors are functions. Infinite differentiability ensures that the derivative maps this space into itself.

Theorem 8.1 (Distinct eigenvalues give independent vectors). If $Tv_j = \lambda_j v_j$, every $v_j \neq 0$, and $\lambda_1, \dots, \lambda_k$ are distinct, then $v_1, \dots, v_k$ are linearly independent.

Proof. Induct on k . A single nonzero vector is independent. From $\sum_{j=1}^k c_j v_j = 0$, applying $T - \lambda_1 I$ gives

$$\sum_{j=2}^k c_j (\lambda_j - \lambda_1) v_j = 0.$$

The induction hypothesis and the nonzero differences give $c_j = 0$ for $j \geq 2$; the original relation then gives $c_1 = 0$. $\square$

In particular, distinct real exponentials $e^{\alpha_j t}$ are linearly independent. Also the sum of eigenspaces for distinct eigenvalues is direct: in $\sum_j w_j = 0$, with $w_j \in E_{\lambda_j}$, discard any zero terms and apply the theorem to the remaining vectors.

2. The characteristic polynomial

For $A \in \mathbb{R}^{n \times n}$, define $p_A(t) = \det(tI - A)$. The coefficient of t^n is 1: only the product of diagonal entries in the determinant expansion can supply n factors of t . Thus p_A has degree exactly n .

Theorem 8.2 (How eigenvalues are found). For a scalar λ , the following are equivalent:

$$\lambda \text{ is an eigenvalue of } A \iff \ker(A - \lambda I) \neq \{0\} \iff p_A(\lambda) = 0.$$

Proof. The first equivalence is the definition. A square matrix has a nonzero kernel precisely when it is not invertible, equivalently when its determinant is zero. Finally $\det(A - \lambda I) = (-1)^n p_A(\lambda)$, so the choice of sign does not change the roots. $\square$

An eigenvalue is found from one scalar equation, but its eigenvectors must still be found by solving the homogeneous system $(A - \lambda I)v = 0$. For a triangular A , $p_A(t) = \prod_i (t - a_{ii})$; its diagonal entries are the eigenvalues, with repetitions.

Example 8.2 (A complete two-dimensional calculation). Let $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$. Then

$$p_A(t) = (t - 1)(t - 3) - 8 = (t - 5)(t + 1).$$

For $\lambda = 5$, the equations are $-4x + 4y = 0$, $2x - 2y = 0$; therefore $E_5 = \text{span}\{(1, 1)^\top\}$. For $\lambda = -1$, they reduce to $x + 2y = 0$; therefore $E_{-1} = \text{span}\{(2, -1)^\top\}$. The two basis vectors have dot product 1, not zero. Distinct eigenvalues give independence without necessarily giving orthogonality.

3. Change of basis and diagonalization

Proposition 8.3 (Similarity). If $B = P^{-1}AP$, then $p_B = p_A$ and $E_\lambda(B) = P^{-1}E_\lambda(A)$.

Proof. Since $tI - B = P^{-1}(tI - A)P$, multiplicativity gives equal determinants. Also $Bz = \lambda z$ if and only if $A(Pz) = \lambda(Pz)$. $\square$

Consequently the characteristic polynomial of a linear map on a finite-dimensional space is independent of the chosen basis.

Theorem 8.4 (Eigenvector basis criterion). A real square matrix A is *diagonalizable over $\mathbb{R}$* if and only if it has a basis of real eigenvectors. In that case

$$P = (v_1 \ \cdots \ v_n), \quad AP = P \text{diag}(\lambda_1, \dots, \lambda_n), \quad P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n).$$

In particular, n distinct eigenvalues give diagonalizability.

Proof. The j -th column of the middle equality is $Av_j = \lambda_j v_j$. If the columns form a basis, P is invertible and the last equality follows. Conversely, a diagonalization provides those n independent eigenvector columns. For distinct eigenvalues, their independence was proved above. $\square$

Example 8.3 (Coordinates and powers). For the preceding A , choose

$$P = \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}, \quad P^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix}.$$

Then $AP = P \text{diag}(5, -1)$. For any integer $k \geq 0$, successive cancellation of $P^{-1}P$ gives

$$A^k = \frac{1}{3} \begin{pmatrix} 5^k + 2(-1)^k & 2 \cdot 5^k - 2(-1)^k \\ 5^k - (-1)^k & 2 \cdot 5^k + (-1)^k \end{pmatrix}.$$

For $k = 2$, this is $\begin{pmatrix} 9 & 16 \\ 8 & 17 \end{pmatrix}$, which direct multiplication confirms. A vector x has eigenbasis coordinates $P^{-1}x$; the map multiplies those coordinates by 5 and -1 .

The *algebraic multiplicity* of λ is its multiplicity as a root of p_A . The *geometric multiplicity* is $\dim E_\lambda(A)$.

Proposition 8.5 (Multiplicity and missing eigenvectors). For each eigenvalue,

$$1 \leq \dim E_\lambda(A) \leq \text{algebraic multiplicity of } \lambda.$$

If the characteristic polynomial splits into linear factors over $\mathbb{R}$, A is diagonalizable exactly when equality holds for every eigenvalue.

Proof. Take a basis of E_λ , of size r , and extend it to a basis of the whole space. In this basis the matrix has block form $\begin{pmatrix} \lambda I_r & B \\ 0 & C \end{pmatrix}$. Its characteristic polynomial is $(t - \lambda)^r \det(tI - C)$, proving the bound. Eigenspaces for distinct eigenvalues have a direct sum. Thus their bases together form a basis of the whole space exactly when their dimensions sum to n . When the polynomial splits, its algebraic multiplicities sum to n ; the individual upper bounds make the stated equality criterion equivalent to that dimension condition. $\square$

Example 8.4 (Repeated roots can leave too few eigenvectors). For

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 4 \end{pmatrix}, \quad p_B(t) = (t - 2)\{(t - 1)(t - 4) + 2\} = (t - 2)^2(t - 3).$$

At $\lambda = 2$, the first row gives $y = 0$ and the second gives $-y - z = 0$, hence $E_2 = \text{span}\{(1, 0, 0)^\top\}$. At $\lambda = 3$, the equations give $y = x$, $z = -2y$, hence $E_3 = \text{span}\{(1, 1, -2)^\top\}$. The eigenspaces together have dimension $2 < 3$, so B is not diagonalizable. By comparison, $\text{diag}(2, 2, 3)$ has the same characteristic polynomial and a three-vector eigenbasis.

Example 8.5 (Back substitution for a triangular matrix). Let

$$C = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 5 & -1 \\ 0 & 0 & 7 \end{pmatrix}.$$

Its eigenvalues are 1, 5, 7. Back substitution gives, respectively,

$$v_1 = (1, 0, 0)^\top, \quad v_5 = (1/4, 1, 0)^\top, \quad v_7 = (1/4, -1/2, 1)^\top.$$

For $\lambda = 7$, the second equation is $-2y - z = 0$; the first is $-6x + y + 2z = 0$. Setting $z = 1$ gives the stated vector. Thus

$$P = \begin{pmatrix} 1 & 1/4 & 1/4 \\ 0 & 1 & -1/2 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1 & -1/4 & -3/8 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}$$

satisfy $P^{-1}CP = \text{diag}(1, 5, 7)$. Here every calculation uses triangular substitution; no general formula for a cubic root is needed.

4. Real matrices can have nonreal eigenvalues

A complex number has the form $a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$. Addition is componentwise, and multiplication gives

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Its conjugate is $\overline{a + bi} = a - bi$, and its absolute value is $|a + bi| = \sqrt{a^2 + b^2}$. Conjugation preserves sums and products. For $a + bi \neq 0$, multiplication by its conjugate gives

$$\frac{1}{a + bi} = \frac{a - bi}{a^2 + b^2}.$$

These arithmetic rules suffice for the calculations below.

For a monic real quadratic $t^2 - bt + c$, completing the square gives

$$t^2 - bt + c = \left(t - \frac{b}{2}\right)^2 - \frac{b^2 - 4c}{4}.$$

If $\Delta = b^2 - 4c \geq 0$, its roots are $(b \pm \sqrt{\Delta})/2$. If $\Delta < 0$, use $(i\sqrt{-\Delta})^2 = \Delta$ to obtain

$$\lambda_{\pm} = \frac{b \pm i\sqrt{-\Delta}}{2}.$$

They are nonreal conjugates. Thus a negative discriminant means no *real* root, not absence of eigenvalues.

For real A , a complex column $z = x + iy$ consists of two real columns x, y . Define $Az = Ax + iAy$. Writing $\lambda = a + bi$, the equation $Az = \lambda z$ is equivalent to the two real equations

$$Ax = ax - by, \quad Ay = bx + ay. \quad (*)$$

A nonzero complex column satisfying them is an eigenvector for λ . The criterion $p_A(\lambda) = 0$ still finds these eigenvalues: elimination and the determinant criterion use only addition, multiplication and division by nonzero numbers, which the arithmetic above permits. The resulting eigenvector must satisfy every row of $(A - \lambda I)z = 0$.

Conjugate pairs. If $Az = \lambda z$, then the real entries of A give

$$A\bar{z} = \overline{Az} = \overline{\lambda z} = \bar{\lambda} \bar{z}, \quad \bar{z} \neq 0.$$

Equivalently, for the real coefficients c_j of p_A ,

$$p_A(\bar{\lambda}) = \sum_j c_j \bar{\lambda}^j = \overline{\sum_j c_j \lambda^j} = \overline{p_A(\lambda)} = 0.$$

Conjugating a factorization $p_A(t) = (t - \lambda)^r q(t)$, $q(\lambda) \neq 0$, shows that $\bar{\lambda}$ has the same multiplicity. When λ is real, (*) gives $Ax = \lambda x$ and $Ay = \lambda y$; at least one is a nonzero real eigenvector. When λ is nonreal, no nonzero *real* eigenvector can have that eigenvalue: the imaginary part of $Av = \lambda v$ would give $0 = bv$.

A real eigenbasis must consist of real vectors. In particular, a real diagonalization requires n independent real eigenvectors. We say that a degree- n polynomial *splits over* $\mathbb{R}$ if it is a product of n real linear factors, with repetitions allowed. A real characteristic polynomial need not have this property.

Proposition 8.6 (Coefficients and eigenvalues). For a real $n \times n$ matrix, the coefficient of t^{n-1} in $p_A(t)$ is $-\text{tr } A$, and $p_A(0) = (-1)^n \det A$. If a factorization $p_A(t) = \prod_{j=1}^n (t - \lambda_j)$ is given, allowing real or nonreal λ_j , then

$$\det A = \prod_{j=1}^n \lambda_j, \quad \text{tr } A = \sum_{j=1}^n \lambda_j.$$

Every real matrix of odd order has at least one real eigenvalue.

Proof. In the permutation expansion of $\det(tI - A)$, the identity permutation contributes $\prod_i (t - a_{ii})$, whose t^{n-1} coefficient is $-\sum_i a_{ii}$. Every other permutation moves at least two indices, so its term has degree at most $n - 2$. Evaluation at zero gives the constant coefficient. Comparing these two coefficients with the stated factorization proves the sum and product formulas. For odd n , the monic polynomial $p_A(t)$ is positive for sufficiently large positive t and negative for sufficiently large negative t . The intermediate value theorem supplies a real root, hence an eigenvalue. $\square$

Example 8.6 (The eigenvalues of a rotation). For

$$R(\theta) = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \quad c = \cos \theta, \quad s = \sin \theta,$$

$$p_R(t) = (t - c)^2 + s^2, \quad \lambda_{\pm} = c \pm is.$$

If $s \neq 0$, these are nonreal and their eigenvectors necessarily have nonreal coordinates. Direct multiplication gives

$$R(\theta) \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} c + is \\ s - ic \end{pmatrix} = (c + is) \begin{pmatrix} 1 \\ -i \end{pmatrix}.$$

Conjugating gives the eigenvector $(1, i)^{\top}$ for $c - is$. There is no real eigenvector, and therefore no real eigenbasis. If $s = 0$, then $R = \pm I$, and every nonzero real vector is an eigenvector with eigenvalue $c = \pm 1$. For a quarter turn, $c = 0, s = 1$, the eigenvalues are $i, -i$: they are purely imaginary, meaning their real parts vanish.

Example 8.7 (Solving a nonreal eigenvalue equation). For the real matrix $A = \begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix}$,

$$p_A(t) = (t - 2)(t - 1) + 3 = t^2 - 3t + 5, \quad \lambda_{\pm} = \frac{3 \pm i\sqrt{11}}{2}.$$

The homogeneous equations are

$$(2 - \lambda)x - y = 0, \quad 3x + (1 - \lambda)y = 0.$$

The first gives $y = (2 - \lambda)x$. Set $x = 1$, giving $z_{\lambda} = (1, 2 - \lambda)^{\top}$. In the second equation,

$$3 + (1 - \lambda)(2 - \lambda) = \lambda^2 - 3\lambda + 5 = 0,$$

so both rows hold. If $x = 0$, the first row forces $y = 0$; hence all eigenvectors for this root are precisely the nonzero complex multiples of z_{λ} . The two displayed eigenvectors are conjugates. There are no real eigenvalues and no real eigenbasis.

Reading: eigenvalues of real orthogonal matrices. For $A^{\top}A = I$, a nonreal eigenvalue still has a conjugate partner because A is real. To find its absolute value, use (*) and real norms:

$$\begin{aligned} \|x\|^2 + \|y\|^2 &= \|Ax\|^2 + \|Ay\|^2 \\ &= \|ax - by\|^2 + \|bx + ay\|^2 \\ &= (a^2 + b^2)(\|x\|^2 + \|y\|^2). \end{aligned}$$

The cross terms cancel. Since $z = x + iy \neq 0$, the final common factor is positive, and $a^2 + b^2 = 1$. Thus $|\lambda| = 1$. A real eigenvalue is consequently 1 or -1 , but the rotation shows why these need not be all the eigenvalues.

5. Diagonal differential equations

If $F'(t) = DF(t)$, $D = \text{diag}(\lambda_1, \dots, \lambda_n)$, then each coordinate solves $f'_j = \lambda_j f_j$. The derivative of $e^{-\lambda_j t} f_j(t)$ is zero, so $f_j(t) = c_j e^{\lambda_j t}$, including when $\lambda_j = 0$. The functions $e^{\lambda_j t} e_j$ form a basis of the solution space: they span by the formula, and evaluation at $t = 0$ proves independence. If $A = PDP^{-1}$, putting $G = P^{-1}F$ gives $G' = DG$, hence

$$F(t) = P \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}) P^{-1} F(0).$$

For $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$ and $F(0) = (1, 0)^{\top}$, $P^{-1}F(0) = (1/3, 1/3)^{\top}$; consequently

$$F(t) = \frac{1}{3} \{ e^{5t} (1, 1)^{\top} + e^{-t} (2, -1)^{\top} \}.$$

Differentiation verifies $F' = AF$, and at zero it gives $(1, 0)^{\top}$. This is the same change of coordinates used for powers of A .

Reading: eigenvectors when cubic roots are implicit

For $A = \begin{pmatrix} 1 & -1 & 2 \\ -2 & 1 & 3 \\ 1 & -1 & 1 \end{pmatrix}$, set $u = t - 1$. Then $p_A(t) = u^3 - u - 1$. Neither 1 nor -1 is a root,

so the rational-root test rules out rational roots. Its derivative is $3u^2 - 1$. At both stationary points, in particular at the local maximum $-1/\sqrt{3}$, the polynomial is negative: the latter value is $2/(3\sqrt{3}) - 1 < 0$. The function therefore has exactly one real root. That root is simple, because neither stationary point is a root.

For that real root u , solve the first two equations with $z = 1$:

$$ux + y = 2, \quad 2x + uy = 3, \quad x = \frac{2u - 3}{u^2 - 2}, \quad y = \frac{3u - 4}{u^2 - 2}.$$

The denominator cannot vanish at a root: $u^2 = 2$ would give $u^3 - u - 1 = u - 1 \neq 0$. The third equation is $x - y - u = 0$; substitution makes its numerator $1 + u - u^3 = 0$. Thus these formulas give a basis vector for the unique real eigenspace. Its dimension is one, so this real matrix has no real eigenbasis. One eigenspace contains infinitely many eigenvectors.

Reading: triangularization with a real-root hypothesis

Suppose p_A splits over $\mathbb{R}$. Then a real change of basis makes A upper triangular. To prove this, induct on the order n . The assertion is immediate for $n = 1$. Choose a real root λ_1 and a nonzero vector $v_1 \in \ker(A - \lambda_1 I)$. Extend v_1 to a real basis, with column matrix P . Then

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & b \\ 0 & B \end{pmatrix}, \quad p_A(t) = (t - \lambda_1)p_B(t).$$

Polynomial cancellation shows that p_B also splits into real factors. By induction there is a real invertible S with $S^{-1}BS$ upper triangular. The change of basis $P \operatorname{diag}(1, S)$ gives

$$\begin{pmatrix} \lambda_1 & bS \\ 0 & S^{-1}BS \end{pmatrix}.$$

Its diagonal entries are the real roots of p_A , with multiplicity. The change of basis is not asserted to be orthogonal. The root hypothesis is essential: the rotation above cannot be similar over $\mathbb{R}$ to a triangular matrix, whose diagonal entries would be real eigenvalues.

Exercises and solutions

A star marks an advanced exercise. Matrices are real. The quadratic, conjugate-pair and skew-symmetric tasks include nonreal eigenvalues and use complex columns when needed; other vector-space and diagonalization tasks retain their stated real scope. Root multiplicities refer to factors of the characteristic polynomial. A complete list of n real eigenvalues is used only when a real factorization has been proved or explicitly assumed.

Scalars, eigenvalues, and eigenspaces

Exercise 8.1 (Three quadratic characteristic equations). Find all eigenvalues, including nonreal ones, of

$$A = \begin{pmatrix} 3 & 5 \\ -2 & -4 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 4 \\ -5 & -5 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 5 \\ 5 & 7 \end{pmatrix}.$$

Solution. Their characteristic polynomials are

$$p_A(t) = (t-3)(t+4) + 10 = (t-1)(t+2),$$

$$p_B(t) = (t-3)(t+5) + 20 = (t+1)^2 + 4,$$

$$p_C(t) = (t-3)(t-7) - 25 = (t-5)^2 - 29.$$

Thus the eigenvalues are $1, -2; -1 + 2i, -1 - 2i$; and $5 + \sqrt{29}, 5 - \sqrt{29}$, respectively.

Exercise 8.2 (Three eigenvalues with different block structures). Find the eigenvalues of

$$A = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 3 & 5 \\ 0 & 5 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

Solution. For A, C , triangularity gives $7, 3, 3$. For B , expansion in the first column gives

$$p_B(t) = (t-7)\{(t-3)(t-4) - 25\} = (t-7)(t^2 - 7t - 13).$$

Its eigenvalues are $7, (7 + \sqrt{101})/2, (7 - \sqrt{101})/2$. The common list for A, C alone does not determine eigenspace dimensions.

Exercise 8.3 (A repeated root and its eigenspace). For $A = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 2 & 1 & 5 \end{pmatrix}$, find the characteristic equation, eigenvalues, rank, and all eigenvectors. Can independent eigenvectors be chosen? Are the eigenspaces orthogonal?

Solution. Expansion in the second row gives $p_A(t) = (t-1)^2(t-5)$. Thus the eigenvalues are $1, 1, 5$, and $\det A = 5$, so its rank is 3. For $\lambda = 1$, the equations are $3y = 0, 2x + y + 4z = 0$; hence $E_1 = \text{span}\{(-2, 0, 1)^\top\}$. For $\lambda = 5$, $-4y = 0$ and $-4x + 3y = 0$, giving $E_5 = \text{span}\{(0, 0, 1)^\top\}$. All eigenvectors are the nonzero vectors in these two lines. One vector from each gives an independent pair, but there is no three-vector eigenbasis. Their dot product is 1, so the lines are not orthogonal.

Exercise 8.4 (The general real two-by-two matrix). For real $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

- Find p_A and all its eigenvalues, including nonreal ones. Show that their sum and product are $\text{tr } A$ and $\det A$.
- Show that if one eigenvalue is nonreal, both are nonreal and one is the conjugate of the other.

(c) If $b = c$, prove that both eigenvalues are real.

Solution. Set $\Delta = (a - d)^2 + 4bc$. Completing the square gives

$$p_A(t) = t^2 - (a + d)t + ad - bc = \left(t - \frac{a + d}{2}\right)^2 - \frac{\Delta}{4}.$$

For $\Delta \geq 0$, the eigenvalues are $(a + d \pm \sqrt{\Delta})/2$, including a repeated root when $\Delta = 0$. For $\Delta < 0$, they are

$$\lambda_{\pm} = \frac{a + d \pm i\sqrt{-\Delta}}{2}.$$

In either case $\lambda_+ + \lambda_- = a + d = \operatorname{tr} A$, and

$$\lambda_+ \lambda_- = \frac{(a + d)^2 - \Delta}{4} = ad - bc = \det A.$$

The negative-discriminant case gives both assertions in (b) explicitly. For $b = c$, $\Delta = (a - d)^2 + 4b^2 \geq 0$, proving (c).

Exercise 8.5 (What does an eigenvalue determine?). (a) Must matrices with the same characteristic polynomial be equal?

(b) Can a nonzero vector be an eigenvector for two distinct eigenvalues of the same matrix?

(c) Can distinct vectors be eigenvectors for the same eigenvalue?

Solution. (a) No: I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have polynomial $(t - 1)^2$, but differ. (b) If $Av = \lambda v = \mu v$, then $(\lambda - \mu)v = 0$; since $v \neq 0$, $\lambda = \mu$. (c) Yes: v and $2v$ have the same eigenvalue whenever v is an eigenvector. For I_2 , the independent vectors e_1, e_2 also share an eigenvalue.

Exercise 8.6 (A shear and a scalar matrix). Find all eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Solution. For A , the only eigenvalue is 1, with algebraic multiplicity two. The equation $(A - I)(x, y)^T = 0$ is $y = 0$; its eigenvectors are $(x, 0)^T$, $x \neq 0$. For B , the only eigenvalue is 2, also of multiplicity two, and every nonzero vector is an eigenvector. Only B has a two-vector eigenbasis.

Exercise 8.7 (Scaling, powers, and inverse). Let $Av = \lambda v$, $v \neq 0$.

(a) Show that tv , $t \neq 0$, has the same eigenvalue, and that v is an eigenvector of tA with eigenvalue $t\lambda$.

(b) Show that v is an eigenvector of $sI - tA$ and of A^k , $k \geq 2$, and give their eigenvalues.

(c) If A is invertible, prove the corresponding assertion for A^{-1} .

Solution. Linearity gives $A(tv) = \lambda(tv)$ and $(tA)v = (t\lambda)v$. Also $(sI - tA)v = (s - t\lambda)v$. Inductively $A^k v = \lambda^k v$. If A is invertible, $\lambda = 0$ would force $v = 0$; thus $\lambda \neq 0$, and multiplying $Av = \lambda v$ by A^{-1} gives $A^{-1}v = \lambda^{-1}v$. The statement about tA allows $t = 0$; every nonzero vector is then an eigenvector of the zero matrix.

Exercise 8.8 (Transpose and eigenspaces). For a real square matrix A , show that A and A^T have the same characteristic polynomial and eigenvalues with multiplicities, including nonreal ones. Must they have the same eigenvectors?

Solution. Transpose invariance of the determinant gives $\det(tI - A^\top) = \det((tI - A)^\top) = \det(tI - A)$. Thus all roots, real or nonreal, and their multiplicities agree. The eigenvectors need not coincide. For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\ker(A - I) = \text{span } e_1$, whereas $\ker(A^\top - I) = \text{span } e_2$.

Exercise 8.9 (Conjugate pairs and odd dimension). (a) Prove that nonreal eigenvalues of a real matrix occur in conjugate pairs, with conjugate eigenvectors.

(b) Prove that every real matrix of odd order has a real eigenvalue.

Solution. For $Az = \lambda z$, $z \neq 0$, conjugate every coordinate:

$$\sum_j a_{ij} \bar{z}_j = \overline{\sum_j a_{ij} z_j} = \overline{\lambda z_i} = \bar{\lambda} \bar{z}_i.$$

Here a_{ij} is real. Thus $A\bar{z} = \bar{\lambda}\bar{z}$, with $\bar{z} \neq 0$; when λ is nonreal, $\bar{\lambda} \neq \lambda$. For(b), the characteristic polynomial is monic of odd degree. For large $M > 0$, $p_A(M) > 0$ and $p_A(-M) < 0$. The intermediate value theorem gives a real λ with $p_A(\lambda) = 0$. Then $A - \lambda I$ has a nonzero real kernel vector. This proves the odd-order assertion without needing a general theorem counting all polynomial roots.

Exercise 8.10 (*Eigenvalues of a real skew-symmetric matrix). Let A be real and $A^\top = -A$. Prove that every eigenvalue is zero or purely imaginary. What follows when the order of A is odd?

Solution. Let $A(x + iy) = (a + bi)(x + iy)$, with real x, y, a, b and $(x, y) \neq (0, 0)$. Equating real and imaginary parts gives

$$Ax = ax - by, \quad Ay = bx + ay.$$

For every real u , skew-symmetry gives $u^\top Au = u^\top A^\top u = -u^\top Au$, hence $u^\top Au = 0$. Take the dot products of the two equations with x, y , then add:

$$0 = x^\top Ax + y^\top Ay = a(\|x\|^2 + \|y\|^2) + b(-x^\top y + y^\top x) = a(\|x\|^2 + \|y\|^2).$$

The factor in parentheses is positive, so $a = 0$. Thus $\lambda = bi$, with $b = 0$ allowed; in particular, the only possible real eigenvalue is zero. In odd order there is a real eigenvalue by the preceding problem. It must be zero, so A is singular.

Exercise 8.11 (Eigenvalues do not count the rank). Let J be $n \times n$, with 1 immediately above the diagonal and zero elsewhere. Find its eigenvalues and rank.

Solution. It is triangular with zero diagonal, so $p_J(t) = t^n$: all eigenvalues are zero. Its action is $J(x_1, \dots, x_n)^\top = (x_2, \dots, x_n, 0)^\top$. Thus its image is $\text{span}(e_1, \dots, e_{n-1})$, of dimension $n - 1$, including rank zero for $n = 1$. Counting nonzero eigenvalues would give the wrong rank for $n > 1$.

Exercise 8.12 (An invertible perturbation). If A is singular, show that some nonzero real ϵ makes $A + \epsilon I$ invertible.

Solution. The polynomial $q(\epsilon) = \det(A + \epsilon I)$ is monic of degree n , so it has only finitely many real roots. Choose any nonzero real number outside that finite set. Then $q(\epsilon) \neq 0$. The choice may be made arbitrarily close to zero, because every punctured interval contains infinitely many real numbers.

Exercise 8.13 (Similarity transports eigenspaces). If $B = P^{-1}AP$, prove that A, B have identical eigenvalues with multiplicities. Must their eigenvectors be identical?

Solution.

$$\det(tI - B) = \det\{P^{-1}(tI - A)P\} = \det(tI - A).$$

The eigenspace relation is $E_\lambda(B) = P^{-1}E_\lambda(A)$, so the actual vectors need not be the same. For example $A = \text{diag}(1, 2)$ and $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ give $B = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$. The eigenvalue 2 has eigenvectors on the line $\text{span } e_2$ for A , but on $\text{span}(-1, 1)^\top$ for B .

Exercise 8.14 (Rectangular products). Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$.

- (a) Show $\det(I_m - AB) = \det(I_n - BA)$.
- (b) Show their nonzero eigenvalues, real or nonreal, agree with algebraic multiplicities.
- (c) If $n > m$, prove that BA has at least $n - m$ zero eigenvalues. Why may there be more?

Solution. Eliminate either off-diagonal block in $\begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix}$ to prove (a). For $t \neq 0$, apply (a) with A/t :

$$t^n p_{AB}(t) = t^m p_{BA}(t).$$

This is a polynomial identity, since it holds for infinitely many t . Away from zero the powers of t do not affect root multiplicities, giving (b). If $n > m$, cancellation gives $p_{BA}(t) = t^{n-m} p_{AB}(t)$, proving (c). There are additional zero roots when AB is singular; for $A = 0$, all n eigenvalues of BA vanish.

Exercise 8.15 (Elementary symmetric functions and power sums). Let $\lambda_1, \lambda_2, \lambda_3$ be real and write

$$\prod_{i=1}^3 (t - \lambda_i) = t^3 + p_2 t^2 + p_1 t + p_0,$$

and put $s_1 = \sum_i \lambda_i$, $s_2 = \lambda_1 \lambda_2 + \lambda_1 \lambda_3 + \lambda_2 \lambda_3$, $s_3 = \lambda_1 \lambda_2 \lambda_3$, and $\sigma_k = \sum_i \lambda_i^k$. Find the p_i in terms of the s_i , and prove that (s_1, s_2, s_3) and $(\sigma_1, \sigma_2, \sigma_3)$ determine one another.

Solution. Expansion gives $p_2 = -s_1$, $p_1 = s_2$, $p_0 = -s_3$. Furthermore

$$\sigma_1 = s_1, \quad \sigma_2 = s_1^2 - 2s_2, \quad \sigma_3 = s_1^3 - 3s_1 s_2 + 3s_3.$$

The first follows by definition, the second by squaring s_1 , and the third by expanding s_1^3 and $s_1 s_2$. Conversely

$$s_1 = \sigma_1, \quad s_2 = (\sigma_1^2 - \sigma_2)/2, \quad s_3 = (\sigma_3 - \sigma_1^3 + 3\sigma_1 s_2)/3.$$

These explicit inverse formulas prove the one-to-one correspondence.

Exercise 8.16 (When traces of powers determine eigenvalues). For real square matrices A, B of order n , assume that both characteristic polynomials split over $\mathbb{R}$. Prove that their eigenvalues agree with algebraic multiplicities if and only if $\text{tr } A^k = \text{tr } B^k$ for $k = 1, \dots, n$.

Solution. Triangularize a matrix as in the reading supplement. Its k -th power has diagonal entries λ_i^k , and trace is unchanged by similarity, since $\text{tr}(XY) = \text{tr}(YX)$. Thus $\text{tr } A^k = \sum_i \lambda_i^k$. To see that the first n power sums determine the characteristic polynomial, write $f(z) = \prod_i (1 - \lambda_i z) = \sum_{r=0}^n (-1)^r e_r z^r$, $e_0 = 1$. The product rule gives $f'(z) = -\sum_i \lambda_i \prod_{j \neq i} (1 - \lambda_j z)$. Multiply the finite geometric identity $(1 - \lambda_i z) \sum_{k=1}^n \lambda_i^k z^{k-1} = \lambda_i - \lambda_i^{n+1} z^n$ by $\prod_{j \neq i} (1 - \lambda_j z)$, and compare coefficients of degrees below n . Summing in i yields the recursion

$$r e_r = \sum_{k=1}^r (-1)^{k-1} e_{r-k} \sigma_k \quad (1 \leq r \leq n).$$

Hence the σ_k determine $e_1, \dots, e_n$ successively. Equal traces give equal characteristic polynomials and therefore equal root multisets. The converse follows directly from the power-sum formula.

Exercise 8.17 (A matrix that cannot be diagonalized). Prove that $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ cannot be diagonalized.

Solution. Its characteristic polynomial is $(t - 1)^3$. The equation $(A - I)x = 0$ gives $x_2 = x_3 = 0$, so its eigenspace has dimension one and cannot provide a basis of $\mathbb{R}^3$. Equivalently, a diagonal matrix with all eigenvalues 1 is I ; similarity to I would make $A = PIP^{-1} = I$, a contradiction.

Exercise 8.18 (Real QR factorization). Let $A \in \mathbb{R}^{m \times n}$ have independent columns, $m \geq n$. Prove $A = QR$, where $Q^\top Q = I_n$ and R is real upper triangular with positive diagonal.

Solution. Apply the real Gram–Schmidt process to the columns a_j . Given $q_1, \dots, q_{j-1}$, put

$$r_{ij} = q_i^\top a_j \quad (i < j), \quad w_j = a_j - \sum_{i < j} q_i r_{ij}.$$

Taking the dot product with any earlier q_h gives $q_h^\top w_j = q_h^\top a_j - r_{hj} = 0$. The vector w_j is nonzero: otherwise a_j would lie in the span of the preceding columns, contrary to independence. Set $r_{jj} = \|w_j\| > 0$, $q_j = w_j/r_{jj}$, and $r_{ij} = 0$ for $i > j$. Induction gives orthonormal columns, and the identities $a_j = \sum_{i \leq j} q_i r_{ij}$ give $A = QR$. All quantities are real by construction.

Exercise 8.19 (A rank-one product and its update). For real $x, y \in \mathbb{R}^n$:

- (a) Prove that $xy^\top$ has $n - 1$ zero eigenvalues and one eigenvalue $y^\top x$, counted with multiplicity.
- (b) Prove $\det(I + xy^\top) = 1 + y^\top x$.
- (c) If $\mu = 1 + y^\top x \neq 0$, prove $(I + xy^\top)^{-1} = I - \mu^{-1}xy^\top$.

Solution. The rectangular determinant identity, applied to an n -by-1 column and a 1-by- n row, gives $p_{xy^\top}(t) = t^{n-1}(t - y^\top x)$, first for $t \neq 0$ and then as a polynomial identity. This proves (a), including $x = 0$, $y = 0$, or $y^\top x = 0$. The same identity at $I + xy^\top$ proves (b). Put $M = xy^\top$; then $M^2 = (y^\top x)M = (\mu - 1)M$. Both products of $I + M$ and $I - \mu^{-1}M$ equal $I + \{1 - \mu^{-1} - \mu^{-1}(\mu - 1)\}M = I$, proving (c).

Exercise 8.20 (Left and right eigenvectors). A real left eigenvector of A for a real λ is a nonzero column y with $y^\top A = \lambda y^\top$.

- (a) Show $A^\top y = \lambda y$.
- (b) If $Ax = \lambda x$, $y^\top A = \mu y^\top$, and $\lambda \neq \mu$, prove $y^\top x = 0$.

Solution. Taking transposes proves (a). For (b), evaluate one scalar in two ways:

$$y^\top Ax = \lambda y^\top x = \mu y^\top x.$$

Thus $(\lambda - \mu)y^\top x = 0$, and the nonzero factor gives $y^\top x = 0$. This pairs a left eigenvector with a right eigenvector; two right eigenvectors need not be orthogonal.

Exercise 8.21 (A companion matrix). Let C have 1 in positions $(j, j + 1)$, $j < n$, last row $(-a_0, -a_1, \dots, -a_{n-1})$, and zero elsewhere. Prove $p_C(t) = t^n + a_{n-1}t^{n-1} + \dots + a_1t + a_0$.

Solution. For $n = 1$, $C = [-a_0]$, giving $t + a_0$. For $n > 1$, expand $tI - C$ in its first column. The top entry contributes t times the determinant of the companion pattern with coefficients $a_1, \dots, a_{n-1}$. The bottom entry a_0 contributes a_0 : its cofactor sign is $(-1)^{n+1}$, and its minor is lower triangular with $n - 1$ diagonal entries -1 , so the signs cancel. Induction yields $t(t^{n-1} + a_{n-1}t^{n-2} + \dots + a_1) + a_0$.

Exercise 8.22 (Commuting matrices and simple eigenvalues). (a) If $AB = BA$ and λ is a simple real eigenvalue of A , prove that each eigenvector of A for λ is an eigenvector of B .

- (b) If A has n distinct real eigenvalues and $AB = BA$, prove that one change of basis diagonalizes both matrices.

Solution. If $Av = \lambda v$, then $A(Bv) = B(Av) = \lambda Bv$. The eigenspace $E_\lambda(A)$ has dimension one because its geometric multiplicity is positive and at most the algebraic multiplicity one. Thus $Bv = \mu v$ for a scalar μ , including $\mu = 0$. Since $v \neq 0$, it is an eigenvector of B . In (b), the distinct eigenvalues of A provide a basis $v_1, \dots, v_n$. By (a), the same vectors are eigenvectors of B ; their column matrix diagonalizes both.

Triangular form and polynomial consequences

Exercise 8.23 (*Strict triangularity and rank decrease). Let A be strictly upper triangular of order n .

- (a) Prove $A^n = 0$.
- (b) Put $r_j = \text{rank}(A^j)$. Prove that $r_{j+1} < r_j$ whenever $r_j > 0$.

Solution. Every nonzero term in $(A^k)_{ij}$ requires a chain $i < i_1 < \dots < i_{k-1} < j$, so $j - i \geq k$. No such pair exists for $k = n$, proving (a). Let $W = \text{im } A^j \neq \{0\}$. Then $AW = \text{im } A^{j+1}$. If the dimensions were equal, the map $A : W \rightarrow W$ would be surjective and hence invertible. Its n -th power would be invertible on the nonzero space W , contradicting $A^n = 0$. Thus $\dim AW < \dim W$, proving (b).

Exercise 8.24 (A growing zero block). Let A, B be upper triangular of order n , $1 \leq k < n$. Assume $a_{ij} = 0$ for $1 \leq i, j \leq k$, and $b_{k+1, k+1} = 0$. Show that AB is upper triangular and its upper-left $(k+1)$ -by- $(k+1)$ block is zero.

Solution. An entry $c_{ij} = \sum_{\ell} a_{i\ell} b_{\ell j}$ can have nonzero terms only for $i \leq \ell \leq j$; hence $c_{ij} = 0$ when $i > j$. If $i \leq j \leq k$, every $a_{i\ell}$ in that range lies in the zero block. If $j = k+1$, the terms with $\ell \leq k$ vanish from A , and the remaining term contains $b_{k+1, k+1} = 0$. This includes $i = k+1$, proving the claim.

Exercise 8.25 (An elementary similarity). Let e_i, e_j be distinct standard basis vectors, and $T(\theta) = I - \theta e_i e_j^\top$.

- (a) Prove $T(\theta)^{-1} = T(-\theta)$.
- (b) If A is upper triangular and $i < j$, prove

$$T(\theta)^{-1} A T(\theta) = A + \theta(e_i e_j^\top A - A e_i e_j^\top),$$

and describe every possibly changed entry.

- (c) If $a_{ii} \neq a_{jj}$, take $\theta = a_{ij}/(a_{ii} - a_{jj})$. What does this achieve?

Solution. Put $E = e_i e_j^\top$. Since $e_j^\top e_i = 0$, $E^2 = 0$; thus $(I + \theta E)(I - \theta E) = I$ in either order. Expansion in (b) gives $A + \theta(EA - AE) - \theta^2 EAE$. Here $EAE = a_{ji} E = 0$, by upper triangularity. For $P = EA - AE$,

$$P_{h\ell} = \delta_{hi} a_{j\ell} - a_{hi} \delta_{j\ell}.$$

The only possible changes are in row i at columns $\ell \geq j$, and in column j at rows $h \leq i$. At their intersection $P_{ij} = a_{jj} - a_{ii}$; hence the chosen θ makes the new (i, j) entry zero. All other changed entries lie farther above the diagonal, and no diagonal entry changes.

Exercise 8.26 (Separate unequal diagonal blocks). Let an upper triangular A have distinct diagonal values $\lambda_1, \dots, \lambda_k$, grouped contiguously with multiplicities $n_1, \dots, n_k$. Write its diagonal blocks as $A_1, \dots, A_k$. Using elementary similarities, prove that A is similar to $\text{diag}(A_1, \dots, A_k)$.

Solution. An entry (i, j) with $i < j$ lying in two different diagonal blocks has $a_{ii} \neq a_{jj}$. Apply the preceding elementary similarity to make it zero. That operation changes only positions (i, ℓ) with $\ell > j$, or (h, j) with $h < i$, besides the targeted entry. These positions have strictly larger distance above the diagonal. They also remain between different diagonal blocks, because $h \leq i < j \leq \ell$ and the blocks are contiguous. Process distances $j - i = 1, 2, \dots, n - 1$, removing all off-block entries at each distance. No operation changes an entry already removed, or any entry within a diagonal block. The resulting matrix is the stated block diagonal matrix; the product of the elementary similarity matrices is invertible.

Exercise 8.27 (A unit triangular eigenvector matrix). If A is upper triangular with distinct diagonal entries, prove that there is a unit upper triangular P such that $P^{-1}AP = \text{diag}(a_{11}, \dots, a_{nn})$.

Solution. For column j , seek an eigenvector v_j for a_{jj} with coordinates $v_{jj} = 1$ and $v_{ij} = 0$ for $i > j$. Rows below j and row j of $(A - a_{jj}I)v_j = 0$ then hold. For $i = j - 1, \dots, 1$, solve successively

$$(a_{ii} - a_{jj})v_{ij} = - \sum_{\ell=i+1}^j a_{i\ell}v_{\ell j}.$$

The divisor is nonzero by distinctness. Thus all v_j exist. Their column matrix P is upper triangular with diagonal 1, hence invertible, and $AP = P \text{diag}(a_{ii})$.

Exercise 8.28 (Triangular inverses). (a) Prove that an upper triangular matrix with nonzero diagonal has an upper triangular inverse with nonzero diagonal.

(b) Prove that a unit upper triangular matrix with integer entries has a unit upper triangular inverse with integer entries.

Solution. In $AX = I$, solve each column by back substitution. For column j , rows $i > j$ give $x_{ij} = 0$, and row j gives $x_{jj} = a_{jj}^{-1}$. The remaining entries follow successively from

$$x_{ij} = -a_{ii}^{-1} \sum_{\ell=i+1}^j a_{i\ell}x_{\ell j} \quad (i < j).$$

This proves (a). In (b), every divisor a_{ii} is 1; the recurrence uses only sums and products of integers, proving (b). Equivalently, if $A = I + N$, strict triangularity gives $N^n = 0$, and $A^{-1} = I - N + \dots + (-N)^{n-1}$.

Exercise 8.29 (Trace of powers with real roots). Let A be real and suppose $p_A(t) = \prod_{j=1}^n (t - \lambda_j)$, with real λ_j . Prove $\text{tr } A^k = \sum_j \lambda_j^k$ for every integer $k \geq 1$.

Solution. The real-root triangularization proof gives $A = PTP^{-1}$, where T is upper triangular with diagonal entries λ_j . Then $A^k = PT^kP^{-1}$, and cyclic trace gives $\text{tr } A^k = \text{tr}(T^kP^{-1}P) = \text{tr } T^k$. For upper triangular U, V , the only possible nonzero term in $(UV)_{jj} = \sum_{\ell} u_{j\ell}v_{\ell j}$ has $\ell = j$. Thus $(UV)_{jj} = u_{jj}v_{jj}$. Induction gives $(T^k)_{jj} = \lambda_j^k$; summing proves the formula.

Exercise 8.30 (Nonzero eigenvalues and rank). If the characteristic polynomial of a real $n \times n$ matrix has r nonzero roots, real or nonreal, counted with algebraic multiplicity, prove that its rank is at least r .

Solution. Put $d = \dim \ker A$. Extending a basis of $\ker A$ to a basis of $\mathbb{R}^n$ represents A by $\begin{pmatrix} 0 & B \\ 0 & C \end{pmatrix}$, where the first diagonal block has order d . Its characteristic polynomial is $t^d \det(tI - C)$. The polynomial on the right after removing t^d has degree $n - d$, so it can have at most $n - d$ nonzero roots with multiplicity. Hence $r \leq n - d = \text{rank } A$. Equality need not hold, as a nonzero strictly upper triangular matrix shows.

Exercise 8.31 (A simple eigenvalue and the converse). (a) If λ is a simple real eigenvalue, prove $\text{rank}(\lambda I - A) = n - 1$. Does the rank condition imply that λ is simple?

(b) Deduce the rank when zero is a simple eigenvalue. If zero has algebraic multiplicity $k \geq 2$, must the rank be $n - k$?

Solution. A simple eigenvalue has a nonzero eigenspace of dimension at most one, hence exactly one. Rank-nullity gives the stated rank. Conversely the rank condition implies a nonzero kernel and thus an eigenvalue, but it need not be simple: $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has rank one and a double eigenvalue zero. If zero is simple, the rank is $n - 1$. For multiplicity $k \geq 2$, take $A = \text{diag}(J_k, I_{n-k})$, where J_k is the shift matrix. Then $p_A(t) = t^k(t - 1)^{n-k}$, but $\text{rank } A = (k - 1) + (n - k) = n - 1 > n - k$.

Exercise 8.32 (*The characteristic equation evaluated at the matrix). For a real $n \times n$ matrix, write $p_A(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_0$. Prove

$$p_A(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_0I = 0.$$

If a factorization $p_A(t) = \prod_j(t - \lambda_j)$ is given, allowing real or nonreal roots, deduce $\prod_j(\lambda_j I - A) = 0$.

Solution. For $n = 1$ the identity is $A - a_{11}I = 0$. For $n > 1$, the entries of $\text{adj}(tI - A)$ are polynomials of degree at most $n - 1$. Write

$$\text{adj}(tI - A) = \sum_{j=0}^{n-1} B_j t^j.$$

The cofactor identity, valid also when $tI - A$ is singular, gives

$$(tI - A) \sum_{j=0}^{n-1} B_j t^j = p_A(t)I.$$

Comparing coefficients yields

$$B_{n-1} = I, \quad B_{k-1} - AB_k = c_k I \quad (1 \leq k < n), \quad -AB_0 = c_0 I.$$

Multiply the middle identity on the left by A^k , sum in k , and include the last identity and $A^n = A^n B_{n-1}$. The result is

$$p_A(A) = A^n B_{n-1} + \sum_{k=1}^{n-1} (A^k B_{k-1} - A^{k+1} B_k) - AB_0 = 0,$$

because every term cancels with its neighbor. No factorization of p_A was needed. When the displayed factorization exists, its factors are polynomials in A , so they commute and their product is $p_A(A)$. Changing the sign of all n factors proves the final assertion.

Further complete calculations and applications

Exercise 8.33 (A reflection). Let $H = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$. Find a nonzero real v with $Hv = v$, and prove that every vector perpendicular to v satisfies $Hw = -w$.

Solution. Write $c = \cos \theta, s = \sin \theta$. If $c \neq 1$, take $v = (s, 1 - c)^\top$, which is nonzero. Multiplication gives $Hv = (cs + s - sc, s^2 - c + c^2)^\top = (s, 1 - c)^\top$. A perpendicular vector is a multiple of $w = -(1 - c, s)^\top$;

$$Hw = (-c + c^2 + s^2, -s + sc - cs)^\top = (1 - c, -s)^\top = -w.$$

If $c = 1$, then $s = 0$, $H = \text{diag}(1, -1)$; take $v = e_1$, and every perpendicular vector is a multiple of e_2 . The two eigendirections describe reflection in the line spanned by v .

Exercise 8.34 (A commuting map preserves an eigenspace). Let $S, T : V \rightarrow V$ be linear and $ST = TS$. If v is an eigenvector of T for λ , show that Sv is also one for λ , provided $Sv \neq 0$.

Solution. $T(Sv) = S(Tv) = S(\lambda v) = \lambda Sv$. The nonzero condition is needed only for the word eigenvector. Without it, the always valid assertion is $S(E_\lambda(T)) \subseteq E_\lambda(T)$.

Exercise 8.35 (Two two-by-two eigenspace calculations). Find the characteristic polynomials, eigenvalues and eigenspace bases of

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 2 \\ -1 & 0 \end{pmatrix}.$$

Solution. For A , $p_A = (t-1)(t-2) - 6 = (t-4)(t+1)$. The first row gives bases $(2, 3)^\top$ for 4 and $(-1, 1)^\top$ for -1 . For B , $p_B = t(t-3) + 2 = (t-1)(t-2)$. The second row gives $x = -\lambda y$, so bases are $(-1, 1)^\top$ for 1 and $(-2, 1)^\top$ for 2. Each listed vector also satisfies the other row. Every root is simple, so each nonzero vector spans its entire eigenspace.

Exercise 8.36 (Three three-by-three eigenspace calculations). Find the characteristic polynomials, eigenvalues and eigenspace bases of

$$A = \begin{pmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{pmatrix}.$$

Solution. For A , expansion in column 2 gives $(t-1)\{(t-4)(t-1) + 2\} = (t-1)(t-2)(t-3)$. Bases for E_1, E_2, E_3 are respectively

$$(0, 1, 0)^\top, \quad (-1, 2, 2)^\top, \quad (-1, 1, 1)^\top.$$

They satisfy $Av = \lambda v$, and simplicity of the roots proves completeness. For B , expansion gives $p_B = (t+2)^2(t-4)$. Here $B + 2I$ has all rows multiples of $(1, -1, 1)$, so

$$E_{-2} = \{(x, y, z)^\top : x - y + z = 0\} = \text{span}\{(1, 1, 0)^\top, (-1, 0, 1)^\top\}.$$

The vector $(1, 1, 2)^\top$ spans E_4 . For C , $p_C = (t-2)^2(t-6)$, and $C - 2I = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} (1, 1, 1)$. Consequently

$$E_2 = \text{span}\{(-1, 1, 0)^\top, (-1, 0, 1)^\top\}, \quad E_6 = \text{span}\{(1, 2, 1)^\top\}.$$

The two independent vectors for each double root exhaust its algebraic multiplicity; the simple eigenspaces are one-dimensional. All three matrices are diagonalizable despite the repeated roots in B, C .

Exercise 8.37 (Two one-dimensional eigenspaces). Find all eigenvalues and eigenspace bases of

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}.$$

Show that the eigenvectors of each matrix span a one-dimensional space.

Solution. For A , $p_A = (t-1)^2$, and the equation $(A-I)(x, y)^\top = 0$ is $x = y$. Thus $E_1 = \text{span}\{(1, 1)^\top\}$. For B , $p_B = (t-2)^2$, and $(B-2I)(x, y)^\top = 0$ is $x = 0$. Thus $E_2 = \text{span}\{(0, 1)^\top\}$. These are all eigenvalues and eigenvectors; both spans have dimension one.

Exercise 8.38 (Real eigenspaces of a cyclic shift and a cubic). Find the real eigenvalues and eigenspace bases of

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & 1 \\ -1 & 3 & 0 \\ -4 & 13 & -1 \end{pmatrix}.$$

Solution. For A , the eigenvector equations give $x_2 = \lambda x_1, x_3 = \lambda^2 x_1, x_4 = \lambda^3 x_1$ and $x_1 = \lambda^4 x_1$. A nonzero solution needs $x_1 \neq 0$. The only real solutions of $\lambda^4 = 1$ are $1, -1$. Their eigenspaces have respective bases $(1, 1, 1, 1)^\top$ and $(1, -1, 1, -1)^\top$.

For B , expansion gives

$$p_B(t) = t^3 - t^2 - t - 2 = (t - 2)(t^2 + t + 1).$$

Since $t^2 + t + 1 = (t + 1/2)^2 + 3/4 > 0$, its only real eigenvalue is 2. Rows two and one of $(B - 2I)(x, y, z)^\top = 0$ give $x = y, z = 3y$. The last row then vanishes, so the eigenspace is $\text{span}\{(1, 1, 3)^\top\}$. Neither matrix has enough real eigenvectors to form a basis.

Exercise 8.39 (Trigonometric differentiation). (a) On the real space $V = \text{span}\{\sin t, \cos t\}$, does differentiation have a nonzero eigenvector?

(b) If k is a nonzero integer, show that $\sin kx, \cos kx$ are eigenvectors of the second derivative and give the eigenvalue.

Solution. In the ordered basis $(\sin t, \cos t)$, the derivative has matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, with polynomial $t^2 + 1$.

Thus it has no real eigenvalue and no nonzero eigenvector in the real space V . Twice differentiating gives $D^2 \sin kx = -k^2 \sin kx$ and $D^2 \cos kx = -k^2 \cos kx$. Both functions are nonzero, and their common eigenvalue is $-k^2$. They may be considered in the invariant space they span, so no domain ambiguity for the second derivative is needed.

Exercise 8.40 (All eigenvectors in a distinct-eigenvalue basis). Suppose $v_1, \dots, v_n$ is an eigenvector basis of T , with distinct eigenvalues $c_1, \dots, c_n$. Prove every eigenvector of T is a scalar multiple of one of these basis vectors.

Solution. Write $v = \sum_i a_i v_i$ and suppose $Tv = \lambda v$. Independence of the basis gives $(c_i - \lambda)a_i = 0$ for every i . At least one $a_i \neq 0$, so $\lambda = c_i$ for that index. Distinctness makes all other coefficients zero. Thus $v = a_i v_i$, with $a_i \neq 0$.

Exercise 8.41 (Three further matrices). Find eigenvalues and eigenspace bases of

$$A = \begin{pmatrix} -1 & 2 & 2 \\ 2 & 2 & 2 \\ -3 & -6 & -6 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{pmatrix}.$$

Solution. Expansion gives

$$p_A = t(t + 2)(t + 3), \quad p_B = (t - 3)(t^2 - 3), \quad p_C = (t - 1)(t - 2)(t - 3).$$

For A , bases for eigenvalues $-3, -2, 0$ are respectively $(-1, 0, 1)^\top, (-2, 1, 0)^\top, (0, -1, 1)^\top$. For B , $E_3 = \text{span } e_1$. For $\lambda = \pm\sqrt{3}$, the last row gives $y = (\lambda + 1)z$; the first gives $x = (2\lambda + 3)z/(\lambda - 3)$. Taking $z = 1$ yields

$$E_\lambda = \text{span} \left\{ \left(\frac{2\lambda + 3}{\lambda - 3}, \lambda + 1, 1 \right)^\top \right\}.$$

The second row holds because $\lambda^2 = 3$. For C , bases for $1, 2, 3$ are respectively $(1, 1, 1)^\top, (2, 3, 3)^\top$, and $(1, 3, 4)^\top$. For instance $C(2, 3, 3)^\top = (4, 6, 6)^\top$. All listed roots are simple; checking the displayed eigenvectors therefore also proves that the bases exhaust the eigenspaces.